Stage 1 — Check Your Understanding

Q1. From drag to permeability

1. Given solid fraction alpha = 0.1 and fibre radius a = 1.5 micrometres = 1.5 x 10^-6 m; the diameter is 3 micrometres.

2. Kuwabara factor: H(alpha) = -(1/2) ln(alpha) - 3/4 + alpha - alpha^2/4.

3. H(0.1) = 1.15129255 - 0.75 + 0.1 - 0.0025 = 0.49879255.

4. Fibre solid volume in a representative volume V is alpha V = pi a^2 ell_f.

5. Fibre length density: n = ell_f/V = alpha/(pi a^2) = 1.41471 x 10^10 m^-2.

6. Drag per unit fibre length: F = 4 pi mu U0 / H(alpha), with units N/m.

7. Pressure-gradient magnitude balances drag per unit volume: Delta p/L = n F = 4 alpha mu U0/[a^2 H(alpha)].

8. Darcy law: Delta p/L = mu U0/k; hence mu U0/k = 4 alpha mu U0/[a^2 H(alpha)].

9. Cancel mu U0: k = a^2 H(alpha)/(4 alpha).

10. k = (1.5 x 10^-6)^2 (0.49879255)/(0.4) = 2.805708 x 10^-12 m^2, or 2.81 x 10^-12 m^2.

The pressure driving force and fibre drag oppose one another; the equation equates their magnitudes. Using the diameter in place of a would incorrectly multiply k by four.

Q2. Why Figure 1’s ordering is not evidence

A three-dimensional isotropic fibre network can have higher permeability than parallel fibres lying across the flow because some fibres are nearly flow-aligned and produce less resistance.

Figure 1 shows that Jackson–James generally predicts higher permeability than the regular-array models, but its crossing of Happel near alpha = 0.28 demonstrates that this ordering is not universal. Because orientation and spatial arrangement change simultaneously between these models, Figure 1 cannot isolate the effect of positional disorder; a controlled comparison must hold solid fraction, diameter distribution, and orientation distribution fixed.

Q3. Most penetrating particle size

Prediction: increasing U0 from 0.1 to 0.2 m/s should shift the most penetrating particle size (MPPS) downward, toward smaller particles.

Pe = d_f U0/D, so Pe is proportional to U0 at fixed particle size.

Diffusion: E_D is proportional to Pe^(-2/3), hence E_D(0.2)/E_D(0.1) = 2^(-2/3) = 0.630 (about a 37% reduction).

Diffusion–interception: E_DR is proportional to Pe^(-1/2), giving a factor 2^(-1/2) = 0.707.

Inertial impaction: Stk = rho_p d_p^2 Cc U0/(18 mu d_f), so Stk and E_I double at fixed particle size under the given fit.

Direct interception E_R has no explicit U0 dependence.

Diffusion dominates small particles and weakens with higher velocity; impaction strengthens for larger particles. The competing changes predict a minimum shifted toward smaller diameters. The uploaded draft reports MPPS = 0.21 micrometres at U0 = 0.1 m/s. Its executable Python code and full numerical settings were not supplied, so the new MPPS cannot be reported as a code-verified numerical result.

Q4. The near-fibre risk

1. Centreline regularized Stokeslet segments: place regularized force segments along the cylinder axis and use a blob satisfying appropriate near-field conditions. This is cheaper for slender fibres, but generic centreline blobs may produce persistent surface-flow errors.

2. Surface-resolved regularized Stokeslets: discretize the cylinder surface with quadrature nodes or triangles and enforce no slip on the actual surface. This is more costly but better suited to near-fibre interception accuracy.

First test: use the surface-resolved method on a single cylinder or small periodic array as a reference solution, then benchmark the lower-cost centreline method against it before scaling up.

Measure leakage by integrating outward normal velocity over the fibre surface:

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Q_leak = integral over S_f of max(0, u dot n) dS.

Lambda_leak = Q_leak/(U0 A_ref).
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Also record max over S_f of |u dot n| and the full no-slip residual. The positive-part integral prevents cancellation between inward and outward errors. Test convergence as discretization spacing and blob width are refined, with a consistently defined reference area A_ref.